

Prenatal Exposure to Noise Stress: Anxiety, Impaired Spatial Memory, and Deteriorated Hippocampal Plasticity in Postnatal Life

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ABSTRACT: Sound pollution is known as an annoying phenomenon in modern life. Especially, development of organisms during fetal life is more sensitive to environmental tensions. To address a link between the behavioral and electrophysiological aspects of brain function with action of hypothalamus-pituitary-adrenal (HPA) axis in stressed animals, this study was carried out on the male Wistar rats prenatally exposed to sound stress. Groups of pregnant rats were exposed to noise stress for 1, 2, and 4 hour(s). The degree of anxiety and the spatial memory were evaluated by elevated plus maze and Morris water maze, respectively. Basic synaptic activity and long-term potentiation (LTP) induction were assessed in the CA3-CA1 pathway of hippocampus. The serum level of corticosterone was measured in the pregnant mothers and the offspring. The behavioral experiments appeared that the stressed animals performed considerably weaker than the control rats. The prenatal stress negatively affected the basic synaptic responses and led to a lower level of LTP. The pregnant animals showed an increased serum corticosterone in comparison with the nonpregnant females. Also the offspring exposed to the noise stress had a more elevated level of corticosterone than the control rats. Our findings indicate that the corticosterone concentration changes markedly coincides the results of behavioral and electrophysiological experiments. We conclude that, similar to other environmental stresses, the sound stress during fetal life efficiently disturbs both cognitive abilities and synaptic activities. The changes in action of HPA axis may contribute to problems of the brain function in the prenatally stress exposed animals. © 2014 Wiley Periodicals, Inc.

KEY WORDS: anxiety; fetal life; learning and memory; sound stress; synaptic plasticity

INTRODUCTION

Experimental researches confirm that during prenatal period the development of an organism is subjected to complex environmental influence (Lordi et al., 1997). Copious animal studies have indicated that maternal stress during pregnancy leave a range of long-term effect

on the offspring (Weinstock, 2001). Although sound is invaluable in communication of most creatures including human beings, however, undesirable, unpleasant, or hurtful sounds are known as noise. Noise is one of the most widespread sources of environmental stress in the world (Wallenius, 2004). It is reported that prenatal exposure to noises negatively influenced the development of fetus as well as neonate (Kjellberg, 1990). Increased ante partum fetal death, congenital anomaly in the central nervous system, impaired social behavior in juvenile stage, and the long-term alteration in the immune function have been attributed to prenatal exposure to noise (Sobrian et al., 1997; Kay et al., 1998). Increased response in hypothalamic–pituitary–adrenal (HPA) axis is also reported in stressful situations (Henry et al., 1994). It is demonstrated that stress-induced secretion of glucocorticoid hormones by adrenal cortex influence cognitive performance (Roozendaal, 2002). Physiological and behavioral modifications under stress are supposed to be depend on developmental actions of glucocorticoids; lipid soluble hormones that may pass from maternal into fetal circulation (Takahashi et al., 1988). Two receptor systems for corticosterone are found in rat brain: mineralocorticoid receptors and glucocorticoid receptors known as Type I and Type II receptors, respectively. As the prenatal brain development is highly sensitive phase of developing brain, the excessive amount of glucocorticoids at critical periods of fetal life can modify neuroendocrine functions and the morphology of several brain structures, including hippocampus in the limbic system (Matthews, 2000).

Hippocampal formation that begins in the prenatal life and continues to the postnatal period (Gould and Tanapat, 1999) is crucial for animals to use spatial information to guide and organize their behavior (Sillers et al., 2003). Evidence indicates that prenatal stresses effectively affect spatial memory (Yaka et al., 2007) as well as synaptic plasticity (Yang et al., 2006) in offspring. The present study was designed to evaluate the effect of prenatal noise stress on the experimental form of synaptic plasticity, spatial representations and anxiety in postnatal period of life.

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MATERIALS AND METHODS

Animals Breeding and Ethics

Virgin female Wistar rats on the night of proestrus were mated with male Wistar rats. The next day, the cage was explored for a vaginal plug, and if found, this day was considered as the first day of pregnancy. Pregnant rats were randomly allocated to one of four (see below) experimental conditions (one animal per cage). Free access to food and water and controlled room conditions (12-h light/dark cycle, light on at 6:00 a.m., temperature 22°C, and air humidity 55–60%) were maintained throughout the experiments. Efforts were made to minimize the number of animals used and their suffering. Adequate measures were taken to minimize animal's pain or discomfort. The experiments were approved by Kashan University of Medical Sciences' Ethical Committee and were carried out in accordance with the Directive 2010/63/EU on the protection of animals used for scientific purposes.

Noise Induction

Broad band traffic noise was recorded in a high-traffic square in capital of Iran (Tehran) by a standard recorder (Panasonic RQ-L11) and amplified by a loud speaker (Power 56W). As level of 95 dB (A) is comparable with the noise level detected in some industrial workplaces the intensity of the sound was set at this intensity by a software (Sonar, Cakewalk). A precision sound level meter (Extech Instruments, MA) was used to monitor the intensity of sound to 95 dB (A) uniformly in the cage. Stress group animals were exposed to the noise (1, 2, and 4 h per day for 7 days) during 08:00–14:00 daily.

Noise Exposure

The rat home cages were placed in a reflective Plexiglas chamber with these dimensions: 60 cm (*w*) × 60 cm (*d*) × 90 cm (*h*). The loud speaker was installed upper left at a distance of 30 cm above the cage.

Pregnant and Nonpregnant Animals

All pregnant rats (except controls) at the third trimester of gestation were exposed daily to the noise stress. Three groups of pregnant animals were received the white noise for 1 (PN1), 2 (PN2), or 4 (PN4) h (s). During this period of time the serum level of corticosterone is the lowest (Dallman et al., 2000). Another pregnant group as control pregnant animals was kept in the noise chamber for the same period of time without noise exposure. The values from the latter group were also compared with a group of nonpregnant female rats.

Offspring

After weaning only the male offspring rats were kept and raised till adulthood. The offspring prenatally exposed to 1, 2, and 4 h noise were assigned to N1, N2, and N4, respectively.

Pups born from the pregnant control rats were assigned as control (CON). The litters at 42–50 days of postnatal age were entered the experiments. Generally, one to two pups were taken from each mother. The N1, N2, N4, and CON rats (one group of each, *n* = 10 in each group) were introduced to the behavioral (elevated plus maze and Morris water maze) testing. Groups of N1, N2, N4, and CON animals were also entered the electrophysiological experiments.

Behavioral Experiments

Spatial memory performance

The spatial memory tests were performed using a Morris water maze (a tank made of galvanized metal, 150 cm diameter, 70 cm depth) filled with water (22°C) up to 20 cm below the rim with visual cues on the surrounding room wall. The pool was divided into four equal quadrants named northeast, southeast, northwest and southwest. A circular platform (10 cm diameter) was submerged 1.5 cm below the water surface and located in the center of one quadrant in constant position throughout the experiment. In the training sessions the rats were given four spatial trials/day for four consecutive days. On each trial, the rat had 90 s to find the hidden platform. If it failed, the animal was guided to the platform and allowed to rest on it for 20 s before being gently dried and moved to a holding cage for the about 10 min intertrial interval. The releasing locations were changed from trial to trial. The probe trials were conducted on day 5, 24 h after the final training session in which the platform was removed, and the rats were released from a fixed starting position and allowed to swim freely for 90 s before being removed. Performance on the training and probe trial phases was measured by escape latency (the time from release of animal in the maze until it located the hidden platform) and percent time (the time spent by the rat in target quadrant where the platform was formerly placed), respectively. All experiments were carried out at night time. During testing, the rat was videotaped from a fixed overhead camera, and custom software (Radiab 7) was used to score performance.

Anxiety evaluation

The elevated plus maze evaluates spatial anxiety in rats (Carobrez and Bertoglio, 2005; Haller and Alicki, 2005). Briefly, the apparatus was elevated 60 cm from the floor and contained four 50 cm × 10 cm arms and a 10 cm × 10 cm center platform. Two opposite arms were enclosed by 40 cm walls, and the other two arms were open. The testing room was moderately illuminated. Each rat was placed in the central platform, facing an open arm and was left to explore the maze for 5 min, while the behavior of animal being observed by the experimenter. As a convention, an entry was defined as having all four paws in the arm. The parameters measured for the plus maze navigation were the number of entries into open arms (OAE) and the time spent in open arms (OAT). The percent of OAE and OAT were calculated as follows:

$$\text{OAE\%} = \frac{\text{number of entries into open arms}}{\text{total number of entries}} \times 100$$

$$\text{OAT\%} = \frac{\text{duration spent in open arms (s)}}{300 \text{ (s)}} \times 100$$

In vivo electrophysiology

The animals were anesthetized with urethane (1.5 g/kg, i.p.) and placed in a head holder in a stereotaxic frame. Two small holes (1 mm diameter) were drilled with a sterile drill bit in the skull to allow the insertion of electrodes in the brain. Electrodes were made from Teflon-coated stainless steel wire (0.008 inch outside diameter) exposed only at the tip (tip separation ~ 0.10 mm). A bipolar stimulating electrode (twisted Teflon-insulated stainless steel) was lowered to the Schaffer collaterals (at the coordinates anteroposterior (AP), 4.2; mediolateral (ML), 3.8 mm from Bregma; dorsoventral (DV), 2.8 mm from the dural surface). A monopolar recording electrode was advanced through the CA1 stratum radiatum (at the coordinates anteroposterior (AP), 3.4; mediolateral (ML), 2.5 mm from Bregma; dorsoventral (DV), 2.5 mm from the dural surface). All coordinates were based on Paxinos and Watson (Paxinos and Watson, 2007) and adapted to the animals according to bregma-lambda distance as previously described (Talaie and Salami, 2013). Extracellular recordings were referenced to an indifferent site on the skull. Constant current stimulation was programmed by a computer software (Neurotrace, WSI, I.R. Iran) and delivered to a stimulator/isolator unit (WPI-A365). Single-pathway recordings of the evoked field extracellular postsynaptic potentials (fEPSPs) were made from the stratum radiatum in the CA1 area of the hippocampus in response to stimulation (two sweeps/min at 30 s intervals) of the ipsilateral Schaffer collateral-commissural pathway. Once the response was stable, an input-output series was recorded using a range of stimulus currents. Then, stimulation intensity was adjusted to elicit a fEPSP slope of 60% of maximum. The baseline fEPSPs were recorded for a 30-min period and averaged for comparison to post-tetanus recordings. Then, long-term potentiation (LTP) was induced by high-frequency stimulation (HFS) of 100 Hz (10 bursts of 10 stimuli, 0.2 ms stimulus duration, and 10 s interburst intervals). Following the tetanic stimulation, recordings continued for at least 2 h. The fEPSPs were amplified on a preamplifier (Electromodule, WSI, I.R. Iran), filtered at 1–3,000 Hz, digitized (10 points/ms) and stored for offline analysis using a software (Potentialize, WSI, I.R. Iran) running on a computer. The data were considered for the changes in slope of the pre- and post-tetanic responses. Data are presented as mean $\pm$ SEM and normalized by taking the pre-tetanus slope of fEPSPs as 100% and comparing the post-tetanus fEPSPs with it. Induction of LTP was defined if mean slope of the fEPSP increased by more than 20% after the HFS train. Blood samples were pooled from the animals when the experiment ended.

Measurement of serum corticosterone concentration

The blood samples for measuring the serum corticosterone concentration were collected from the animal groups entered the

experiments and mothers either exposed or unexposed to the sound stress. The blood samples were taken from: all pregnant groups, the nonpregnant female group, the offspring entered behavioral experiments before and after the testing, and the offspring used in electrophysiological experiments after recordings finished. Animals were deeply anesthetized and blood was collected from heart. The pooled blood was then collected into a sterile tube (Kisker, Germany). The blood samples were centrifuged at 2,500 rpm for 10 min. The serum was collected and frozen at -20°C until assayed. The serum corticosterone concentration was measured using a commercially available radioimmunoassay (RIA) kit (DRG, Germany) according to the manufacturer's instruction. The manufacturer supplied protocol was implemented. In brief, corticosterone level in serum samples was single-determined via interpolation from a standard curve derived from six calibrators (ranging from 25 to 1,500 ng/ml).

Data Analysis

One-way analysis of variance (ANOVA) with corticosterone as between-subjects factor and noise exposure as within-subjects factor was performed on the corticosterone level of serum. Also one-way ANOVA with learning the water maze task, remembering the learned task and the anxiety evaluation as between-subjects factors and noise exposure as within-subjects factor was carried out on the behavioral procedures. Two-way ANOVA with change in the slope of fEPSPs as between-subjects factor and noise exposure as within-subjects factor was applied to the electrophysiological experiments; however, the potentiation was interpreted as LTP if the slope of fEPSP was increased at least 20% post-tetanus. LSD was applied as post hoc test. Data are expressed as mean $\pm$ SEM. The basis of statistical decision was a significance Alpha level of $P < 0.05$.

RESULTS

The Effect of Prenatal Sound Stress on Learning the Spatial Maze Task

The spatial learning and memory of the animals prenatally exposed to noise stress were evaluated via two strategies of maze navigation; training and probe trial.

Training

Analysis of variance indicated that the four groups of animals introduced to the training phase displayed a different performance in learning the maze task ($F_{(3,156)} = 6.542$; $P < 0.0001$). All groups improved their maze steering over 4 days of experiment, however, the post hoc test appeared that the N2 ($P = 0.028$) and N4 ($P < 0.0001$) groups behaved significantly lower compared with their control counterparts. One hour exposure to the noise not considerably affected the maze performance. Figure 1A demonstrates how the different groups of animals learned the maze task.

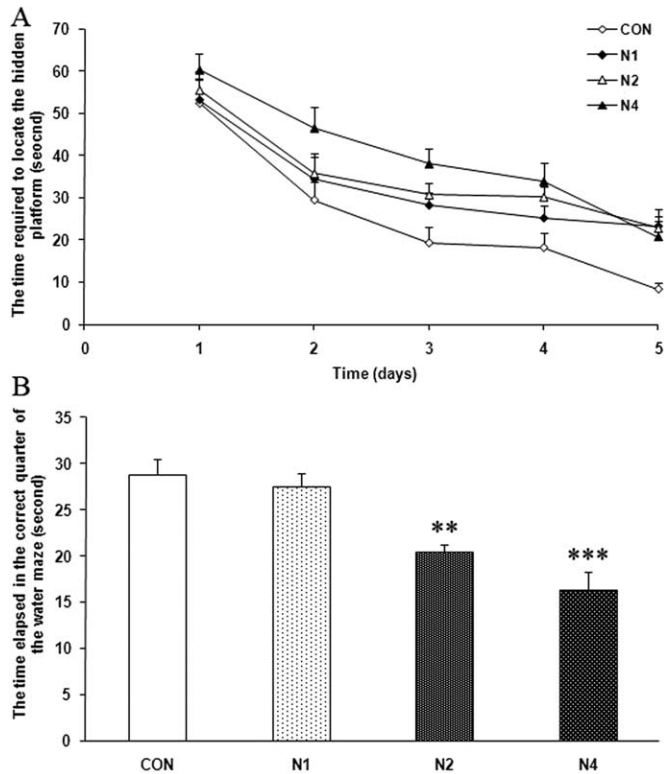


FIGURE 1. Performance of the different groups of animals in the Morris water maze. A) The time required to locate the hidden platform in the water maze during the learning stage. The animals exposed to the noise stress for 2 ($P = 0.028$) and 4 h ($P < 0.0001$) during the fetal life performed significantly lower than the controls. B) The time elapsed in the correct quarter of the water maze during the retrieval test. The groups receiving the noise stress for 2 (** $P = 0.002$) and 4 h (** $P < 0.0001$) during the prenatal life displayed a weaker performance than the control animals. Data are shown as mean $\pm$ SEM.

Probe trial

Remembering the learned task was assessed by removing the platform and measuring elapsing time in the correct quadrant of maze. Statistical analysis appeared a general variation between testing groups in performing the retrieval task ($F_{(3,36)} = 15.491$; $P < 0.0001$). Consistent to the findings in the learning phase, the N2 ($P = 0.002$) and N4 ($P < 0.0001$) groups had a shorter navigation in the correct quadrant when compared with the CON rats. The values are 16.4 ± 1.83 s and 20.4 ± 0.83 s for the N2 and N4 groups, respectively. Behavior of the animals in N1 group resembled that in the CON one so that a marked difference was evident between the N1 rats with the N2 ($P < 0.011$) and N4 ($P < 0.0001$) groups (Fig. 1B).

The Effect of Prenatal Sound Stress on Anxiety

To evaluate the degree of anxiety in the animals prenatally exposed to noise stress they were introduced to a plus maze task. Entry to the open arms and the elapsing time in the entered arms were two factors by which the level of anxiety

was measured. ANOVA indicated a variation between the performance of the different groups of animals ($F_{(3,36)} = 7.368$; $P < 0.0001$). The offspring born from mothers exposed to 2 ($P = 0.005$) and 4 ($P = 0.004$) h noise stress displayed a significant decrease in entering the open arms in comparison to their control counterparts (Fig. 2A). In addition, the animals in the N2 ($P = 0.002$) and N4 ($P = 0.0001$) groups passed a shorter time in the open arms. The N1 group had an almost similar function as the CON animals in both entering the open arms and the duration of stay (Fig. 2B).

The Serum Concentration of Corticosterone in the Offspring Under the Behavioral Testing

The corticosterone concentration in serum of the animals performing the Morris water maze tasks was assessed as the function of HPA axis. ANOVA indicated a general statistical difference between the serum level of corticosterone in the four groups of rats before (Fig. 3A) and after (Fig. 3B) the behavioral testing ($F_{(7,72)} = 50.877$; $P < 0.0001$). The results of corticosterone measurement before performing the maze testing

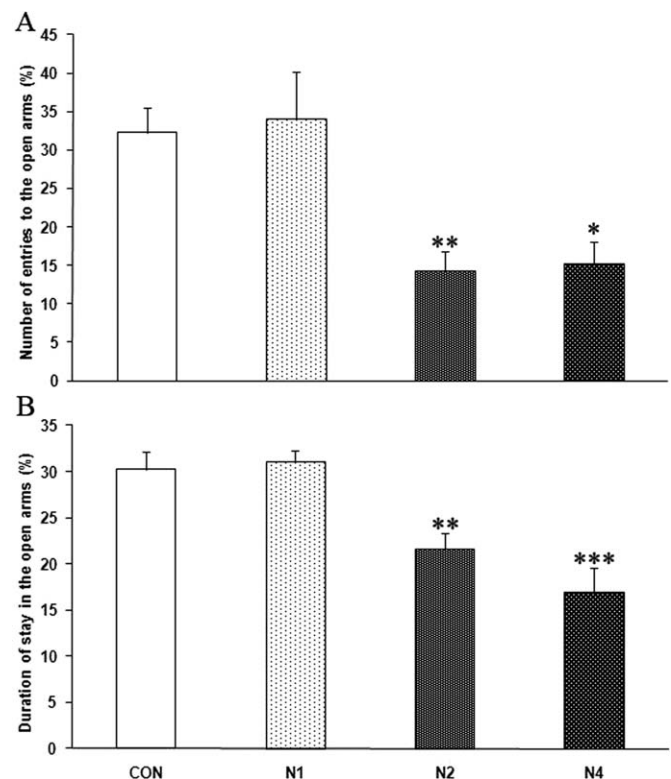


FIGURE 2. The elevated plus maze navigation by the different groups of animals. A) Percentage of the entered open arms by the different groups of rats during the plus maze searching. While the noise stress led a marked decrease in entering the open arms in the N2 (** $P = 0.005$) and N4 (* $P = 0.004$) groups it had no effect on behavior of the animals in the N1 rats. B) Percentage of the duration of stay in the entered open arms during the plus maze navigation. The N2 (** $P < 0.002$) and N4 (** $P < 0.0001$) animals displayed a considerable decrease in the time of the open arm steering. The N1 animals performed the same as the CON rats. Data are shown as mean $\pm$ SEM.

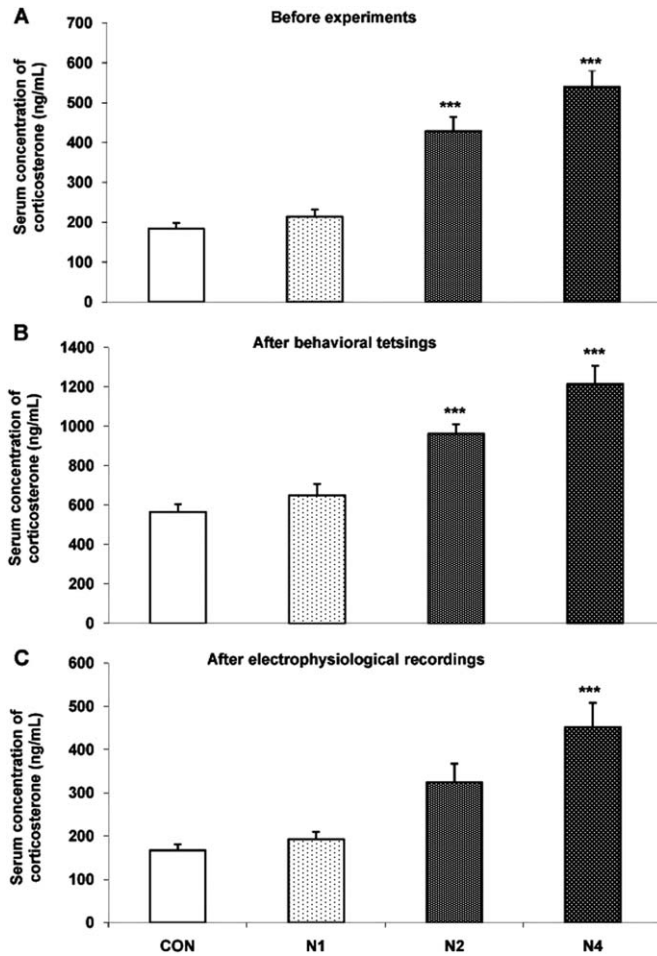


FIGURE 3. The serum level of corticosterone in the pregnant and nonpregnant mothers and, offspring. **A)** The corticosterone concentration in the offspring before the behavioral trials. The prenatal noise stress led to a surge of corticosterone in the serum of both the N2 ($***P < 0.0001$) and N4 ($***P < 0.0001$) groups. The concentration of serum corticosterone in the N1 rats resembled the CON animals. **B)** The maze performance itself led to a significant increase in the serum level of corticosterone in all testing groups ($***P < 0.0001$). Also a substantial variation was evident between the N2 and N4 groups compared with the CON animals ($***P < 0.0001$). **C)** The concentration of corticosterone in the offspring after the electrophysiological recording session. While comparison of the values between the CON animals with the N1 and N2 groups appeared no noticeable variation the abrupt surge of corticosterone in the N4 rats display a substantial difference with their CON counterparts ($***P < 0.0001$). Data are shown as mean $\pm$ SEM.

indicated that the prenatal exposure to the sound stress itself causes an abrupt rise in the serum corticosterone. The values for the N2 (428.68 ± 36.18 ng/ml) and N4 (539.77 ± 42.41 ng/ml) groups are highly different ($P < 0.0001$) than the CON rats (183.71 ± 14.65 ng/ml). The serum level of corticosterone in the N1 (213.93 ± 18.01 ng/ml) animals showed no significant change compared with the control group (Figure 3A). The maze navigation further increased the corticosterone hormone where, compared with the premaze testing values, the serum concentration of corticosterone was enormously

increased in the CON (565.69 ± 39.33 ng/ml, $P < 0.0001$), N1 (649.51 ± 58.27 ng/ml, $P < 0.0001$), N2 (961.97 ± 48.79 ng/ml, $P < 0.0001$), and N4 ($1,215.12 \pm 92.37$ ng/ml, $P < 0.0001$) groups. Measurement of the corticosterone level after the behavioral experiments indicated a substantial variation between the N2 and N4 groups with the CON animals ($P < 0.0001$). Figure 3B compares the serum level of corticosterone after the morris water maze performance.

The Serum Concentration of Corticosterone in the Offspring Under the Electrophysiological Recordings

The serum level of corticosterone was also concerned when the electrophysiological experiments finished. The data analysis appeared that the concentration of the hormone was significantly different between the four groups ($F_{(3,36)} = 12.724$; $P < 0.0001$). The post hoc test indicated that the variation observed between the CON rats (167.06 ± 14.16 ng/ml) and the animals prenatally exposed to the noise stress for 1 (192.84 ± 17.17 ng/ml) and 2 (324.75 ± 42.88 ng/ml) h was not statistically significant. Exposure to the sound stress for 4 h abruptly increased the serum corticosterone to 452.51 ± 55.86 ng/ml which was considerably different compared with the CON group ($P < 0.0001$). We found no noticeable difference between the corticosterone level of serum in the N1, N2, and N4 groups. Figure 3C shows how the noise exposure affected secretion of the noise-induced hormone in the rats under the electrophysiological experiments.

The Serum Concentration of Corticosterone in the Male, Pregnant, and Nonpregnant Animals

To assess if exposure to the noise stress also underlies the pregnant rats themselves the concentration of corticosterone was measured in the four pregnant groups and the nonpregnant group (as control). Data analysis indicated that the differences between the five groups of animals were significant ($F_{(4,45)} = 29.493$; $P < 0.0001$). Comparison between the control pregnant (451.18 ± 38.84 ng/ml) and nonpregnant female (270.28 ± 21.38 ng/ml) groups showed that the pregnancy itself elevates the serum level of corticosterone ($P = 0.003$). Substantial variations were also evident between the control pregnant with the PN2 (659.40 ± 19.21 ng/ml; $P < 0.0001$) and PN4 (720.62 ± 26.54 ng/ml; $P < 0.0001$) groups. The serum level of corticosterone in the PN1 animals (497.53 ± 48.53 ng/ml) resembled that in the control pregnant group. The variations between the PN1 with the PN2 ($P < 0.011$) and PN4 ($P < 0.0001$) rats were statistically significant. Figure 4 demonstrates measurement of corticosterone in serum of the different female groups.

Electrophysiological Recordings in the CA1 Area of Hippocampus

In the electrophysiological experiments the effect of noise on the pre- and post-tetanus fEPSPs in the hippocampus of the

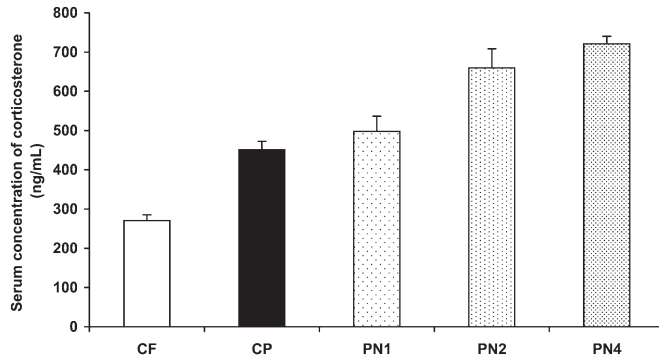


FIGURE 4. The effect of pregnancy on the serum concentration of corticosterone. The control pregnant (CP) rats showed an elevated level of serum corticosterone compared with the nonpregnant control female (CF) group (** $P = 0.003$). Also the serum level of corticosterone was significantly higher in the PN2 and PN4 groups when compared with the CP animals (** $P < 0.0001$). Data are shown as mean $\pm$ SEM.

four groups of offspring was evaluated. ANOVA assessment appeared that the difference between slope of the field recordings was statistically significant ($F_{(7,2492)} = 71.288$, $P < 0.0001$).

The effect of noise stress on the baseline responses

The basic synaptic activity showed to be prominently affected by exposing to prenatal sound stress. Mean slope of the evoked fEPSPs recorded in the CA1 area of hippocampus of the CON rats was 480.53 ± 17.31 mV/ms. The slope of the pre-HFS responses displayed a significant decrease in the N1 (377.02 ± 9.99 mV/ms), N2 (382.05 ± 6.74 mV/ms), and

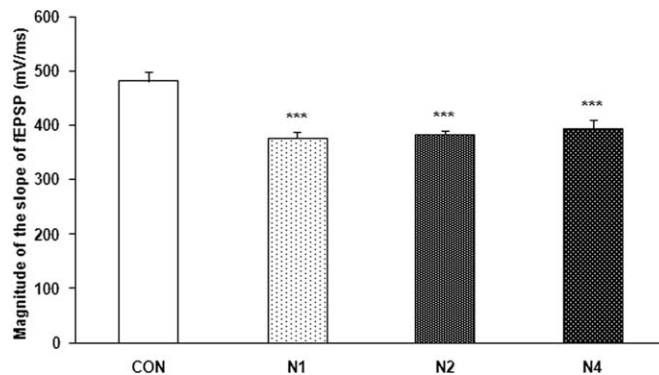


FIGURE 5. Histograms show the slope of the baseline fEPSPs taken from CA1 field of the hippocampi of the control and prenatally noise exposed animals. The comparisons between the N1, N2, and N4 groups with the CON rats indicates that the noise exposure during the fetal life pronouncedly affect the basic synaptic transmission in the hippocampal neuronal circuits (** $P < 0.0001$). No difference was evident between the basic responses recorded in the N1, N2, and N4 groups. Data are shown as mean $\pm$ SEM.

Hippocampus

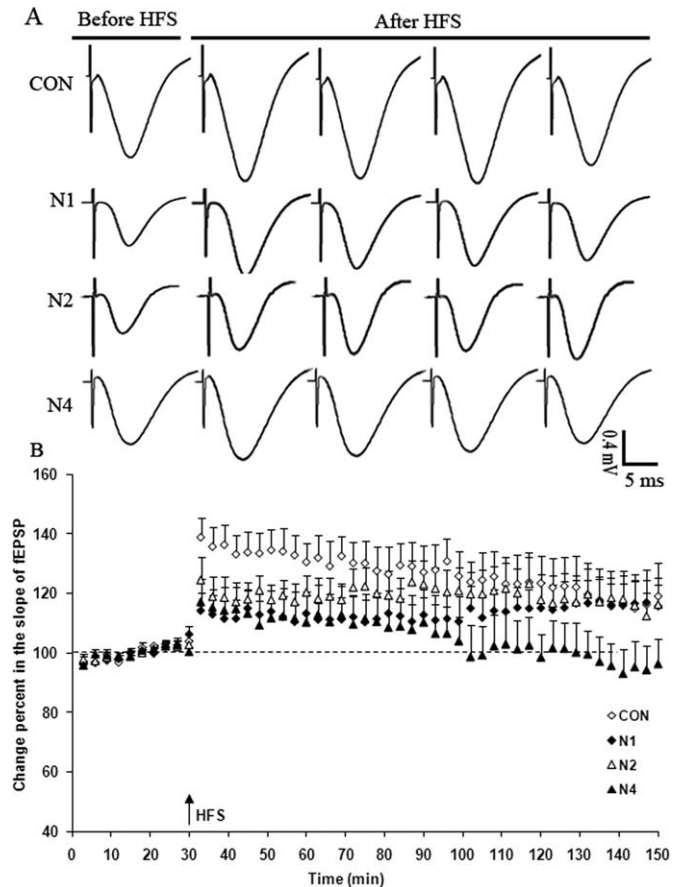


FIGURE 6. Induction of LTP in the fEPSPs recorded in the CA1 area after tetanization of the CA3–CA1 pathway. A) Representative traces illustrating changes in the slope of field potentials recorded in the CA1 area 30 min before and 30, 60, 90, and 120 min after application of the tetanic stimulation. B) Time course of change in the slope of pre- and post-tetanus fEPSPs. The first 30 min of experiment shows the baseline recordings. Tetanization induced a substantial LTP in the fEPSPs recorded in the CON rats ($P < 0.0001$). The potentiation observed in the noise stress exposed groups was less than the criteria determined for LTP in this study and significantly smaller compared with the CON group ($P < 0.0001$). Arrow indicates the time of high frequency stimulation (HFS). Each point shows average data obtained during 3 min.

N4 (398.69 ± 15.54 mV/ms) animals compared with their CON counterparts ($P < 0.0001$). There was a negligible difference between the slope of the baseline recordings in the N1, N2, and N4. Figure 5 demonstrates the slope values of the basic synaptic transmission.

The effect of noise stress on induction of LTP

Tetanzing the Schaffer collateral–CA1 pathway elicited a noticeable LTP in the fEPSPs recorded in the hippocampus of CON animals (from 480.53 ± 17.31 mV/ms to 584.67 ± 8.14 mV/ms, $P < 0.0001$); more than 21% increase in the slope of the fEPSPs. Although the tetanized fEPSPs in the N1 (14.18%), N2 (18.24%), and N4 (2.33%) appeared a potentiation, however, the post-tetanus enhancements did not fulfill

the criterion of LTP occurrence in this study. The level of potentiation in the CON group was statistically higher than the stressed groups ($P < 0.0001$). Figure 6 illustrates the time course of the slope changes of fEPSPs over electrophysiological recordings.

DISCUSSION

Copious reports have indicated an altered behavioral performance in offspring born from mothers under stress during a critical period of gestation. The cognitive changes are proved to be associated with electrophysiological modifications in some parts of brain. This study evaluates the effect of sound stress in fetal life on spatial memory, HPA axis function, and characteristics of synaptic transmission in the postnatal period of life.

Behavioral Considerations

Our results demonstrated that the prenatal noise exposure effectively weaken the behavioral performances during the postnatal life. The negative effect of the sound stress was more pronounced in the N2 and N4 groups. The animals in the N2 and N4 groups showed to perform reasonably lower than the CON group in either plus maze navigation or Morris water maze tasks. Anxiety of the animals was reflected in decreased entering the open arms of plus maze and passing a shorter time in these arms. Also we found that the stressed animals swim a longer path for locating the hidden platform in the water maze and spend a shorter navigation time in the target quadrant; indicating an impaired spatial learning and memory. Animal studies have demonstrated that maternal stress during pregnancy give rise to a variety of long-term effect in offspring including anxiety, learning impairments and reduced attention (Igosheva et al., 2007). Some kinds of stress are reported to efficiently impair spatial learning and memory abilities of the offspring (Yaka et al., 2007). Acute stress disrupts both spatial and recognition memory as assessed by a number of behavioral tasks including radial arm maze, water maze, radial arm water maze, and spontaneous recognition memory tests (Cazakoff et al., 2010). Nishio et al. demonstrated that prenatal stress exposure obviously influences the learning abilities in the offspring (Nishio et al., 2001). Prenatally stressed rats spend a shorter time in the open arms of plus maze; demonstrating a higher degree of anxiety (Murmu et al., 2006). Emotional and cognitive deficits are reported in the offspring born from mothers exposed to stress during gestation (Talge et al., 2007). Beside the studies confirming our results some findings disagree with the present work are also reported. Several researchers found that prenatal stress either did not influence or decreased learning performance in the adult offspring (Lordi et al., 1997). This discrepancy in results could be due to some factors including the task used, the stressor, the time and duration of

stress application and some characteristics of the animals used in different studies such as sex and age (Joels, 2006).

Electrophysiological Considerations

Our results indicated that the sound pollution profoundly affects the basic synaptic transmission. The noise stress decreased the slope of pre-tetanus fEPSPs. Also, in contrast to the control animals, the post-HFS responses recorded in all groups prenatally exposed to the noise stress could not fulfill the criterion of LTP induction in this study. Several reports have demonstrated an impaired LTP and facilitated LTD in hippocampal CA1 area in the prenatal stress exposed offspring (Son et al., 2006; Yang et al., 2006). Yaka et al. reported a diminished LTP as well as impaired spatial learning in the animals introduced to a series of prenatal stresses (Yaka et al., 2007). Contrary to the studies just mentioned Fujioka et al. reported that short-lasting and mild prenatal stress (30 min daily restraining) promotes learning and, improves LTP and neurogenesis in hippocampus of adult offspring. On the other hand, they found that long-lasting and severe prenatal stress lead to reverse results (Fujioka et al., 2006).

Hypothalamus-Pituitary-Adrenal Axis Function

Measurement of serum concentration of corticosterone appeared that pregnancy itself highly increases secretion of the hormone. Also the mothers under the noise stress and the offspring at the N2 and N4 groups had a higher serum level of the glucocorticoid. These findings indicate that the noise pollution efficiently increases activity of the HPA axis in both mothers and pups. On the other hand, the maze navigations elevated the serum corticosterone in all groups of rats whether exposed or unexposed to the stress. However, the hormone surge was significantly higher in the stressed rats compared with the CON animals. The augmentation in the serum level of the glucocorticoid was also observed in the animals entered the electrophysiological experiments; however, the changes were much less than in the behavioral testing and were statistically significant only in the N4 group. This could be rational as the awaked animals under behavioral testing are much more stressful than the anesthetized ones during the electrophysiological recordings. Taken together, the behavioral performances and the invasive in vivo experiments strengthen activity of the HPA axis. From these considerations it can be concluded that the anxiety and memory behavioral results from the maze performances coincides the synaptic transmission findings from the electrophysiological recordings in that the higher the level of corticosterone the lower the capability of cognitive function and, basic and tetanized synaptic activity.

Concurrently, some human studies prove the glucocorticoid hormone alterations during both pregnancy and fetal period. It is demonstrated that maternal stress increases level of corticosterone either in mother or fetuses (Takahashi et al., 1998). However, cortisol levels rise in fetal serum remain about 13-fold lower than in the mother (Gitau et al., 2001), because about 80% is metabolized by 11β -hydroxysteroid dehydrogenase

(11 β -HSD) to inactive cortisone in the placenta (Weinstock, 2008). Stress possibly reduces activity of 11 β -HSD as well as cortisol binding globulin (Weinstock, 2008) resulting in exposure of the fetus to excessive amounts of the steroid (Van den Bergh et al., 2005). Both types of the glucocorticoid receptors are expressed in hippocampus (Zhang et al., 2013). It is reported that stress hormones like glucocorticoid could interact with their receptors in the fetal brain and influence the neuronal differentiation and function at different stages of development (Weinstock, 2008). Henry et al. demonstrated that both type of glucocorticoid receptors are down regulated in hippocampus of the animals exposed to stress during prenatal life (Henry et al., 1994). Down regulation of the hippocampal corticosteroid receptors (Weinstock, 2005), increased HPA responsiveness, and stress-induced increases in glucocorticoids during gestation might be reasons of changes in behavior of the stressed offspring (Takahashi et al., 1998). Some studies Alfarez et al. showed that, similar to stress conditions, corticosterone itself impairs hippocampal LTP (Alfarez et al., 2002). In contrast, Diamond et al. demonstrated that low to moderate levels of corticosterone enhance LTP. Moreover, they found that lack of the hormone diminishes LTP occurrence (Diamond et al., 1992). If corticosterone dose dependently affect production of hippocampal LTP (Joels, 2006), needs further investigation.

Probable Mechanisms Influencing Parts of Brain Involved in Learning and Memory and LTP

It has been shown that stress exerts an intense impression on molecular mechanisms underlying synaptic plasticity (Howland and Wang, 2008). Both the hippocampal LTP and the hippocampal related spatial memory are known to be subject to plasticity of synaptic transmission. The protocol of tetanization and the recruited hippocampal circuit in the present study expect a NMDA receptor dependent LTP. Positive, particularly calcium, ions influx through the NMDA receptors channel founds cellular and molecular changes in the pre and postsynaptic neurons that leads to long lasting alterations in capabilities of the synapse (Molnar, 2011). Changes in structure of NMDA receptors including the combination of subunits might affect their key function in the synaptic plasticity (Paoletti and Neyton, 2007). Son et al. found that offspring of mice bearing chronic immobilization stress during pregnancy efficiently reduce the expression of NR1 and NR2B subunits of NMDA receptor at the hippocampal synapses. Evidence indicates that reduced expression of the NR2B subunits decreases ionic currents through NMDA receptor channel (Son et al., 2006). The change in conductance of the receptor channel may affect the NMDA receptor dependent LTP and, thus, hippocampus dependent spatial memory (Son et al., 2006). Gerges et al showed that stress diminishes expression of phosphorylated calcium/calmodulin-dependent protein kinase II in hippocampus (Gerges et al., 2004). These proteins are known to involve synaptic plasticity. There are evidences demonstrating that prenatal stress perturbs biosynthesis of brain-derived neurotro-

phic factor (BDNF) in the hippocampus (Van den Hove et al., 2006). BDNF is necessary for formation of the synapses and synaptic plasticity as well (Cohen-Cory et al., 2010). From these considerations it can be concluded that, through affecting the expression or activation of some key molecular agents, the prenatal stress might has a negative impact on the hippocampal LTP and spatial memory.

The Stress-Induced Changes in Structure of Hippocampus

The structural characteristics of the pathways involved in synaptic plasticity and learning and memory phenomena are also subjected to stress induced alterations. As hippocampus is rich in glucocorticoid receptors (Zhang et al., 2013) the adrenal hormones produced by stress could affect structure of hippocampus. Accordingly, chronic exposure to either stress or stress-induced glucocorticoids change structure of hippocampus in different aspects including neurochemistry, excitability, neurogenesis, neuronal morphology and cell death (for review see (Conrad, 2010)). Coe et al. reported a reduced volume of hippocampus in nonhuman offspring prenatally exposed to unpredictable noise (Coe et al., 2003). Concomitantly, Morales-Medina et al. showed that 3 weeks administration of corticosterone in rats causes dendritic remodeling and spine density reorganization in pyramidal neurons of the CA1 area of the hippocampus (Morales-Medina et al., 2009). Also in a histological study Bock et al. showed that prenatal stress exposure give rise to shrinkage of dendrites in either CA3 or CA1 areas of the hippocampus (Bock et al., 2011). Therefore, stress or stress-induced hormone corticosterone may affect LTP and, learning and memory through some alterations in organization of the hippocampus.

The Inhibitory Role of Amygdala

Amygdala as a part of limbic system anatomically projects to several hippocampal regions including the CA1 area (Yilmazer-Hanke, 2008) and involve in stress behaviors and memory modulation. Stress is activating molecular plasticity in amygdala (Ilin and Richter-Levin, 2009) and, amygdalo-hippocampal projections exert an inhibitory impact on synaptic plasticity in hippocampus (Akirav and Richter-Levin, 2006). Lesion to the amygdala inhibits stress-induced deficits of hippocampal LTP and hippocampus mediated spatial memory in rats (Kim et al., 2001). Moreover, application of GABA_A receptor agonist into amygdala prevents stress-induced physiological and behavioral results (Kim et al., 2005). In line, hippocampus of stressed animals receiving intra-amygdalar infusion of musimol displays a substantial LTP comparable with control animals (Kim et al., 2005). Thus, a declined LTP and cognitive function in the stressed animals might be due to a stress activated inhibitory impact of the amygdala on the hippocampus.

The Role of Oxidative Stressor

Oxidative stressors may contribute to stress-induced learning and memory impairments (Sharma et al., 2009). Oxidative

stress is shown to induce oxidative damage of the hippocampus which is associated with a pronounced decrease in glucocorticoid receptors in the CA1 area of hippocampus (Kobayashi et al., 2009). It is suggested that, through damage in the HPA axis, oxidative stressors increase glucocorticoids levels and, thus, impairs cognitive functions (Ghadroost et al., 2011). As demonstrated by several researchers glucocorticoids or chronic stress exposure may lead to oxidative injury of hippocampus; resulting in impairments of learning and memory (Sato et al., 2010).

CONCLUSIONS

Taken together, the prenatal noise stress endorses anxiety and deeply undergoes the hippocampus linked learning and memory processes at postnatal age. Also the sound pollution during the fetal life pronouncedly influences the basic synaptic activity as well as synaptic plasticity in the hippocampal circuits of the offspring. Our findings are agreed to previous studies assessing the effects of other stress protocols on cognitive functions and synaptic specificities. Contribution of glucocorticoids to various possible mechanisms involved in stress-induced problems suggests that the prenatal sound stress may affect the brain function more pronouncedly through reregulation of the HPA axis action. Hence, considering a link between the sound pollution and the HPA axis, future research could be focused on the cognitive functions and its electrophysiological mechanisms. However, impact of the other contributions, including some neurotransmitter projections to the hippocampus, to deleterious effect of sound stress on fetal neurodevelopment need to be considered.

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